

## **CIRCUITLY.IN**

### **Obstacle Avoidance Robot DIY Kit**

#### **User Manual**

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#### **Introduction**

Thank you for purchasing the Circuitly.in Obstacle Avoidance Robot DIY Kit.

This exciting STEM learning kit helps students learn about Arduino programming, robotics, sensors, motors, and autonomous navigation. The robot uses an HC-SR04 Ultrasonic Sensor mounted on an SG90 Servo Motor to detect obstacles and automatically change direction.

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#### **Learning Objectives**

By building this project, students will learn:

- Arduino programming basics
  - Ultrasonic distance measurement
  - Servo motor control
  - DC motor control using L298N
  - Autonomous robot navigation
  - Electronics assembly and wiring
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#### **Kit Contents**

| <b>Item</b>               | <b>Quantity</b> |
|---------------------------|-----------------|
| Arduino UNO               | 1               |
| L298N Motor Driver        | 1               |
| HC-SR04 Ultrasonic Sensor | 1               |
| SG90 Servo Motor          | 1               |

| Item                       | Quantity |
|----------------------------|----------|
| BO Motors                  | 2        |
| Robot Chassis              | 1        |
| Wheels                     | 2        |
| Castor Wheel               | 1        |
| Battery Holder with Switch | 1        |
| Jumper Wires               | Assorted |
| Mounting Hardware          | Set      |

### Power Connections

1. Connect battery positive (+) to L298N 12V terminal.
2. Connect battery negative (-) to L298N GND terminal.
3. Connect battery output to Arduino DC Jack.
4. Ensure all grounds are connected together.

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### Assembly Instructions

#### Step 1: Assemble Chassis

- Fix both BO motors to the chassis.
- Install wheels.
- Attach castor wheel.

#### Step 2: Mount Electronics

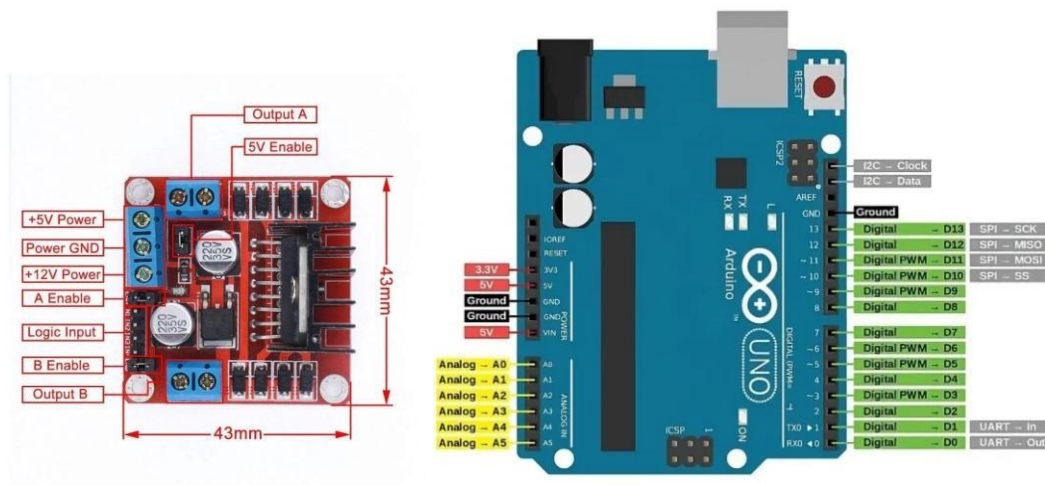
- Mount Arduino UNO.
- Mount L298N Motor Driver.
- Secure battery holder.

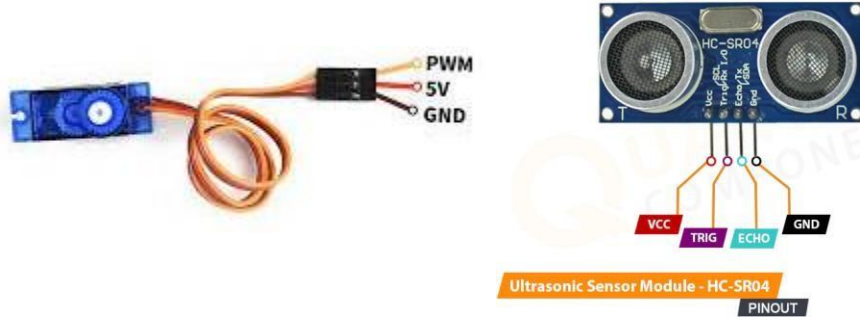
### Step 3: Install Ultrasonic Sensor

- Attach HC-SR04 sensor to SG90 servo bracket.
- Fix servo bracket on the front side of the robot.

### Step 4: Connection Pin out

|                          |                     |
|--------------------------|---------------------|
| <b>L298N driver Pins</b> | <b>Arduino Pins</b> |
| ENA                      | 5                   |
| IN1                      | 2                   |
| IN2                      | 3                   |
| IN3                      | 7                   |
| IN4                      | 8                   |
| ENB                      | 6                   |
|                          |                     |
| <b>Ultrasonic Sensor</b> | <b>Arduino Pins</b> |
| VCC                      | 5V                  |
| GND                      | GND                 |
| TRIG                     | A0                  |
| ECHO                     | A1                  |
|                          |                     |
| <b>SG90</b>              | <b>Arduino Pins</b> |
| VCC                      | 5V                  |
| GND                      | GND                 |
| Signal                   | 10                  |





### Step 5: Upload Code

- Connect Arduino to computer.
  - You can download Arduino ide from <https://support.arduino.cc/hc/en-us/articles/360019833020-Download-and-install-Arduino-IDE#installation-instructions>
  - Open Arduino IDE.
  - You can download code from <https://github.com/circuitlyin/SmartTrack-kit>
  - Watch the step by step guide video <https://www.youtube.com/@Circuitlyin>
  - Select Board: Arduino UNO.
  - Select correct COM Port.
  - Upload the provided code.
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### How It Works

1. Robot moves forward.
2. Ultrasonic sensor continuously measures distance.
3. When obstacle is detected:
  - Robot stops.
  - Servo rotates sensor left and right.
  - Distances are compared.
4. Robot turns toward the side with more free space.
5. Robot continues moving.

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## Operating Instructions

1. Place robot on flat surface.
2. Turn ON power switch.
3. Robot will start scanning surroundings.
4. It will automatically avoid obstacles.

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## Safety Precautions

- Do not short circuit battery terminals.
- Keep robot away from water.
- Do not force wheels by hand while powered.
- Disconnect battery when not in use.
- Adult supervision is recommended for young children.

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## Troubleshooting

### Robot Does Not Move

Check:

- Battery charge
- Power switch
- Motor wiring
- Arduino code upload

### robot keep turning in circles and not going straight

Motor speeds are unequal — use `analogWrite` with different PWM values for each motor to compensate. Or The HC-SR04 is getting false echoes from the chassis itself. Make sure the sensor is elevated above the chassis by at least 2–3cm on a bracket.

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## Ultrasonic Sensor Not Detecting Objects

Check:

- TRIG connected to A0
- ECHO connected to A1

- Sensor VCC and GND
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### **Servo Not Rotating**

Check:

- Signal wire connected to D10
  - Servo power connection
  - Battery voltage
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### **Motors Moving in Wrong Direction**

Swap motor wires on L298N output terminals.

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### **Technical Specifications**

- Operating Voltage: 7V–12V DC
  - Controller: Arduino UNO
  - Distance Sensor: HC-SR04
  - Servo Motor: SG90
  - Motor Driver: L298N
  - Drive Motors: 2 x BO Motors
  - Programming Language: Arduino C/C++
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### **Support**

For tutorials, code updates, and support:

Website: [www.circuitly.in](http://www.circuitly.in)

Email: [support@circuitly.in](mailto:support@circuitly.in)

Happy Learning and Building!

Team Circuitly.in

